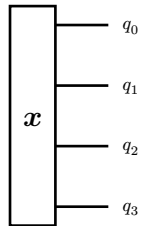
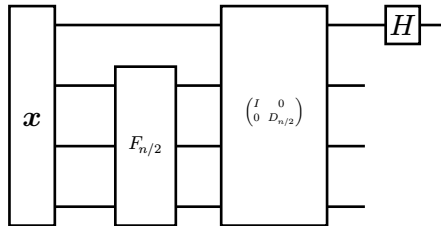


(a) Re-index



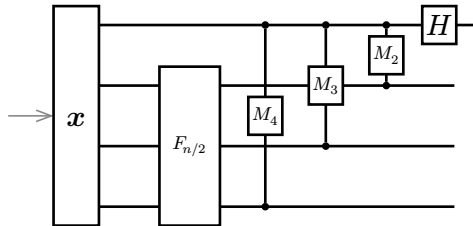
x as rank- k tensor

(b) Recurse



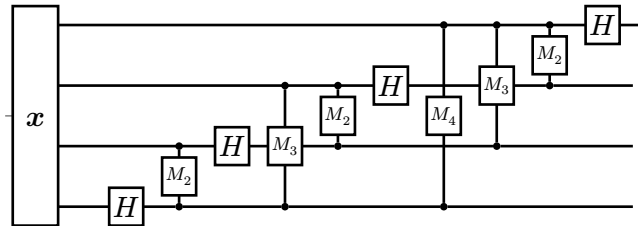
one butterfly layer

(c) Decompose



twiddle $\rightarrow M_j$ chain

(d) Relax



$H \in U(2)$, $M_j \in U(1)^4$